

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Embedded System (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Give a Proper definition of ES. Classify ES based on Performance and Functional Requirements, Deterministic behavior, and triggering. Additionally, list and define the three primary characteristics of ES that distinguish it from other computing systems. 7
- b) Design a home automation system to open the door of your room when you are close to your door using proximity sensor and motors. Explain the key components in your system design. 8
2. a) Explain the process involved in interrupt handling in AVR. 7
- b) Write a program in AVR microcontroller to create a square wave of 50% duty cycle on the PORTB.5 bit. (Use Timer0 to generate time delay.) 8

OR

Write a program in ATmega32 to receive bytes of data serially and out them on Port A with the baud rate of 9600, 8 – bit data and 1 stop bit.

3. a) In a multithreaded embedded system, what problems can arise if shared resources like I/O ports or memory buffers are not properly synchronized? Provide a real-world example (e.g., automotive control system). 7

OR

What does a real-time embedded system mean? How does RTOS ensure fairness in scheduling? Discuss trade-offs between fairness, latency, and complexity in task synchronization.

- b) Explain Context Switching process in RTOS.
4. a) Define VHDL. Explain sub program and packages in VHDL with examples.

- b) Mention different data types in VHDL. Write VHDL code for a Full Adder using behavioral modeling style. Also, verify it by drawing a wave. 8
5. a) Design an embedded Smart Lock System using a keypad, fingerprint sensor, and OLED display. Explain how SPI is used for the fingerprint module, I²C for the display, and UART for GSM/Bluetooth communication. 7
- b) A smart farming solution includes soil moisture sensors, weather stations, and irrigation controllers connected via an embedded system. Data should be transmitted to a central monitoring unit and accessible via mobile apps 8
- i. Compare LoraWAN and Zigbee for long-range, low-power communication in an agricultural setting.
- ii. What are the trade-offs between wired and wireless communication in this scenario?
6. a) Explain how you can interface a temperature sensor with AVR microcontroller with its assembly program. 7
- b) For a Smart Home Automation System 8
- i. Describe how Arduino can be used to develop a home automation system.
- ii. What sensors and actuators are required for this system?
- iii. How does Arduino handle the communication between devices?
- iv. What are the limitations of using Arduino for a large-scale home automation project?
7. Write short notes on: (**Any two**) 2×5
- a) Multithreading vs multitasking
- b) Analog and digital sensors
- c) Embedded systems and IoT